24CS2506 – Building Applications using GPT-4

Experiment 5: Automated Legal Document Analyzer using GPT-4

1. What is a Legal Document?

A **legal document** is a written document that defines legal rights, responsibilities, obligations, agreements, or procedures.

Examples

- Employment Agreement
- Rental Agreement
- Non-Disclosure Agreement (NDA)
- Service Agreement
- Sales Agreement
- Loan Agreement
- Privacy Policy
- Terms and Conditions
- Business Contract
- Legal Notice

2. What is Legal Document Analysis?

Legal document analysis is the process of examining a legal document to identify and understand important information such as:

- Parties involved
- Purpose of the agreement
- Obligations
- Rights
- Payment terms
- Duration
- Termination conditions
- Confidentiality requirements
- Important dates
- Governing law
- Potentially important clauses

An AI-based system can help users understand lengthy legal documents more quickly.

3. Why is Legal Document Analysis Difficult?

Legal documents can be difficult to understand because they often contain:

- Complex terminology
- Long sentences
- Formal language
- Cross-references
- Multiple clauses

- Exceptions
- Conditions
- Legal terminology
- Large amounts of text

Example

A contract may contain several pages describing:

payment, termination, confidentiality, liability, intellectual property and dispute resolution.

Manually identifying all these details can be time-consuming.

4. What is Legal Document Summarization?

Legal document summarization is the process of generating a shorter version of a legal document while preserving its important information.

Input

```
Long legal document
        ↓
     GPT-4
        ↓
Concise summary
```

Example

Original:

The employee shall maintain strict confidentiality regarding all proprietary information obtained during the period of employment and shall not disclose such information to any third party without prior written authorization...

Summary

The employee must keep company confidential information private and cannot disclose it without authorization.

5. Components of a Good Legal Analysis Prompt

A useful prompt should specify:

Role

```
You are a legal document analysis assistant.
```

Task

```
Analyze and summarize the provided document.
```

Information to Extract

```
Identify:
- Parties
- Purpose
- Obligations
- Payment terms
- Duration
- Termination
- Important dates
```

Constraint

```
Use only information present in the document.
Do not invent facts.
```

Output Format

```
Present the result using headings and bullet points.
```

6. Important Information in a Contract

A basic analyzer can identify the following:

Information	Example
Parties	Company A and Company B
Purpose	Software development
Duration	12 months
Payment	₹50,000/month
Obligations	Deliver software according to specifications
Termination	30 days written notice
Confidentiality	Information must remain confidential
Governing Law	Applicable jurisdiction
Important Dates	Start and end dates

7. Key Clause Identification

The system can identify important clauses such as:

Confidentiality

Information that must not be disclosed.

Payment

Details about fees, invoices and payment schedules.

Termination

Conditions under which the agreement can end.

Liability

Responsibilities relating to losses or damages.

Intellectual Property

Ownership and usage rights relating to created materials.

Dispute Resolution

How disagreements between parties should be resolved.

8. Extractive vs Abstractive Summarization

Extractive Summarization

Selects important sentences directly from the original document.

```
Original Document
       ↓
Important Sentences
       ↓
Summary
```

Abstractive Summarization

Generates a new summary using the meaning of the original document.

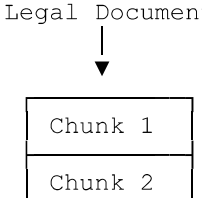
```
Original Document
       ↓
Understand Meaning
       ↓
Generate New Text
       ↓
Summary
```

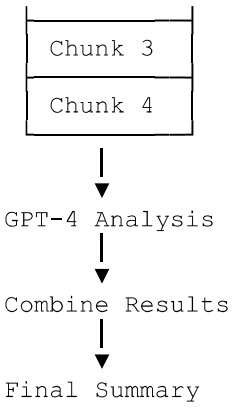
GPT-4 can be used for **abstractive summarization**.

9. Chunking Long Documents

Legal documents can be very long.

Instead of sending the entire document at once, it can be divided into smaller sections.





For a beginner laboratory, students can initially work with a **short text document** and then extend the application to chunking.

Recommended Output Format

The system can generate:

```
LEGAL DOCUMENT ANALYSIS
--------------------------------

Document Type:
Employment Agreement

Parties:
Employer and Employee

Purpose:
Employment of the individual by the organization.

Duration:
12 months

Key Obligations:
• Employee must maintain confidentiality.
• Employee must perform assigned duties.

Payment:
Monthly salary as specified in the agreement.

Termination:
30 days written notice.

Important Dates:
Start Date: 01-07-2026

Potentially Important Clauses:
• Confidentiality
• Termination
• Intellectual Property

Summary:
```

This agreement defines the terms of employment...

Suggested System Workflow

